



CompuCell3D model of cell migration reproduces chemotaxis

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ABSTRACT

Chemotaxis combines three processes: directional sensing, polarity reorientation and migration. Directed migration plays an important role in immune response, metastasis, wound healing and development. To describe chemotaxis, we extend a previously computational model of a 3D single cell, that presents three compartments (lamellipodium, nucleus and cytoplasm), whose migration on a flat surface quantitatively describes experiments. The simulation is built in the framework of CompuCell3D, an environment based on the Cellular Potts Model. In our extension, we treat chemotaxis as a compound process rather than a response to a potential force. We propose robust protocols to measure cell persistence, drift speed, terminal speed, chemotactic efficiency, taxis time, and we analyze cell migration dynamics in the cell reference frame from position and polarization recordings through time. Our metrics can be applied to experimental results and allow quantitative comparison between simulations and experiments. We found that our simulated cells exhibit a trade-off between polarization stability and chemotactic efficiency. Specifically, we found that cells with lower protrusion forces and smaller lamellipodia exhibit an increased ability to undergo chemotaxis. We also noticed no significant change in cell movement due to external chemical gradient when analyzing cell displacement in the cell reference frame. Our results demonstrate the importance of measuring cell polarity throughout the entire cell trajectory, and treating velocity quantities carefully when cell movement is diffusive at short time intervals. The simulation we developed is adequate to the development of new measurement protocols, and it helps paving the way to more complex multicellular simulations to model collective migration and their interaction with external fields, which are under development on this date.

1. Introduction

By sensing environmental cues, migrating cells can follow gradients of temperature, light intensity, electric or chemical fields, of substrate roughness and stiffness [1]. This ability is determinant for cell survival and large scale organization in multicellular systems. In fact, embryogenesis, inflammatory response, wound healing, and metastasis depend on the coordination between cells and their environment. Understanding how such phenomena take place, in either computational, experimental or theoretical scopes, requires the consideration of robust measurement protocols and model's faithfulness to observed migration mechanisms. We start by covering general observations and methods of single cell migration to support our work on chemotaxis.

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One way to characterize cell movement of isolated mesenchymal cells on isotropic flat surfaces is to calculate Mean Squared Displacement (MSD) as a function of the time interval used to measure cell displacement. MSD curves allow to identify ballistic and diffusive regimes in different time scales. The Fürth equation [2] has been used to describe single cell migration kinetics for over a century. It describes cell's MSD with two kinetic regimes: a ballistic regime for short time intervals and a diffusive regime for long time intervals. This equation is, in fact, identical to the solution of the Langevin problem for the velocity of a passive point particle moving in a viscous fluid. In this case, the particle movement at short time intervals is ballistic, therefore instantaneous velocity is a well defined quantity. Opposite to this paradigm, Thomas and collaborators [3] analyzed trajectory data from experiments of a cell migrating on flat surfaces and found three different regimes. For very short time intervals, the movement is diffusive, for intermediary time intervals the movement is ballistic-like, and for long time scales the movement is diffusive again. Diffusive behavior for short time intervals poses a problem in describing cell movement using instantaneous velocity, since the ratio of cell displacement and time interval diverges when the time interval goes to zero.

To deal with the short time scale diffusive regime, Thomas and collaborators [3] empirically added a term, linear in Δt , to the Fürth equation, resulting in what they call the modified Fürth equation. This new equation allowed the proposition of natural units for a universal family of curves: cells kinetics only differ by the duration of the short time diffusion interval relative to the onset of the long time interval diffusion. They have successfully fit cells' kinetics for 12 different experimental set-ups from 5 laboratories.

To understand how cells can persistently move while being diffusive in short time scales, we need first to consider cells as active, extensive, irregular and polarized bodies, for which Langevin equations do not apply. Their energy is not harvested from medium thermal activity, being rather based on ATP processing by a complex, internal machinery, where constant polymerization and depolymerization of actin network takes place, often showing a preferential axis. Cell's center of mass position is the measured quantity to produce MSD curves and to estimate cell speed. The cell front, or lamellipodium, is the main cellular structure responsible for the thrust in cell migration. It is very thin compared to cell body and creates many short lived protrusions in different directions. Each of these protrusions and fluctuations happen simultaneously, at least considering the smallest time intervals available to mesoscopic measurements (of the order of microns). For reviews on cell migration, check Refs. [4–6]. It is reasonable to consider that these fluctuations provide a source of noise to the center of mass position, explaining the short time diffusion observed in single cell migration experiments.

The Langevin model cannot be directly applied to a system that presents diffusion at short time intervals, since in this case instantaneous velocity is not a measurable quantity. To circumvent this, de Almeida and collaborators [7] proposed a theoretical, stochastic model of a particle that presents an internally defined polarization. The dynamics of the center of mass position follows a Langevin process in the direction of the polarization and a Wiener process in the polarization's orthogonal direction. The authors have succeeded in obtaining the modified Fürth equation for the MSD (used in Refs. [3,8]), with the short time interval diffusion term. More recently, Giardini and collaborators [9] analytically solved a system of stochastic differential equations with isotropic noise, resembling the behavior of out simulation in CC3D, and they successfully reproduced the velocity probability density functions observed in this paper.

In 2020, Fortuna and collaborators [8] developed a simulation model of single cell migration that quantitatively reproduces experimental data using Cellular Potts Model (CPM) [10–12] in CompuCell3D [13]. This model shows cell spontaneous polarization, spontaneous polarization reorientation and migration in the polarization direction. The resulting MSD curves are well fit by the modified Fürth equation, and Mean Velocity Correlation Functions (mVACF) reproduce the experimental results. The fitting procedure, using the modified Fürth equation, allowed a translation between laboratory and simulation units. In fact, the model serves as a proxy to the experimental cell behavior and provides a powerful tool to investigate cell migration kinetics, to be verified later in experiments. Thomas et al. [14] investigated the polarization definition in this model, verified that the dynamics of cell migration is different in the polarization direction and orthogonal direction, and proposed a polarization measurement procedure that also applies to experiments.

Here we present a modification of Fortuna and collaborator's model to simulate chemotactic response, and we provide an appropriate MSD equation to characterize cell kinetics in this condition. We aim at a chemotactic response mechanism focused on migration reorientation, agreeing with experimental evidence that external chemical gradients act primarily over polarization orientation rather than cell speed [15,16]. Then we further investigate cell polarization and the diffusive regime for short time scales to propose new robust metrics. Finally, we apply both new and standard metrics to characterize the response and compare it to the non chemotactic, isotropic case.

The paper is organized as follows. In Section 2 we present the simulation model, discuss how we implement the response to external gradients, and propose adequate measures to investigate both the cellular response and the short-time-interval behavior with and without external fields. In Section 3 we present the results and in Section 4 we discuss and conclude with new ideas for experiments.

2. Methods

2.1. Simulation model

2.1.1. Model structure

Fortuna and collaborators proposed a simulation model that describes single cell migration on flat substrates [8]. The model is based on CPM [10–12] and is built in the CompuCell3D (CC3D) environment [13]. The simulation consists in a three-dimensional grid containing the migrating cell immersed in a Medium and laying over a 2D flat Substrate. The simulated cell contains three

compartments: Lamel, Nuc, and Cyto. We list the simulation objects, their properties, behaviors, interactions, and dynamics below. We indicate the new features that we added to the previous model [8] to allow chemotaxis. The supplemental materials offer more details on how the model works [17].

- **Medium, representing the cell environment**

- **Properties:** none.
- **Behaviors:** none.
- **Interactions:** Interacts via contact energy with other CPM objects. Participates in the protrusion mechanism of Lamel, where Lamel advances towards Medium voxels according to the gradient of F-actin.
- **Dynamics:** CPM dynamics.

- **Substrate, representing the 2D substrate where the cell adheres and migrates**

- **Properties:** none.
- **Behaviors:** none.
- **Interactions:** Interacts via contact energy with other CPM objects.
- **Dynamics:** No dynamics, this cell type stays frozen.

- **Nuc, representing Nucleus**

- **Properties:** volume and compressibility.
- **Behaviors:** intrinsic CPM fluctuations.
- **Interactions:** Interacts via contact energy with other CPM objects. The way in which contact energies work in CPM is such that a high contact energy between two objects makes them avoid each other. In particular, Nuc has high energy with all CPM objects except Cyto, so Nuc stays inside the Cyto compartment.
- **Dynamics:** CPM dynamics.

- **Cyto, representing the cytoplasm, excluding the lamellipodium**

- **Properties:** volume and compressibility.
- **Behaviors:** intrinsic CPM fluctuations.
- **Behaviors (NEW):** Senses the concentration of chemical fields across the base voxels (voxels that belong to the Cyto compartment and are located in the xy plane of $z = 1$).
- **Interactions:** Interacts via contact energy with other CPM objects.
- **Interactions (NEW):** Nucleates Lamel voxels at the base, preferentially where the concentration of the chemical field is high.
- **Dynamics:** CPM dynamics.
- **Dynamics (NEW):** Can lose voxels of its base due to the dynamics of Lamel nucleation, see Eq. (2).

- **Lamel, representing the lamellipodium**

- **Properties:** volume and compressibility.
- **Behaviors:** intrinsic CPM fluctuations. Protrudes towards Medium voxels, following negative gradients of the F-actin field.
- **Behaviors (NEW):** Senses the concentration of chemical fields across the voxels of its base.
- **Interactions:** Interacts via contact energy with other CPM objects. In particular, Lamel has very low contact energy with the Substrate, making it spread and flatten.
- **Interactions (NEW):** Can be nucleated at the base of Cyto, according to the concentration of the external chemical field, see Eq. (2).
- **Dynamics:** CPM dynamics, including the protrusion force that is exclusive of the Lamel compartment.
- **Dynamics (NEW):** Voxels are created in the base of Cyto due to the mechanism of Lamel nucleation, see Eq. (2).

- **F-actin Field, representing the actin network in the lamellipodium**

- **Properties:** intensity (binary, either 0 or 1).
- **Behaviors:** none.
- **Interactions:** Interacts with Lamel, causing Lamel to protrude towards Medium.
- **Dynamics:** assumes value 1 in Lamel voxels that are in contact with the Substrate and assumes value zero everywhere else. This dynamics produces a negative gradient from Lamel voxels to Medium voxels, causing Lamel to advance towards Medium, according to Eq. (1).

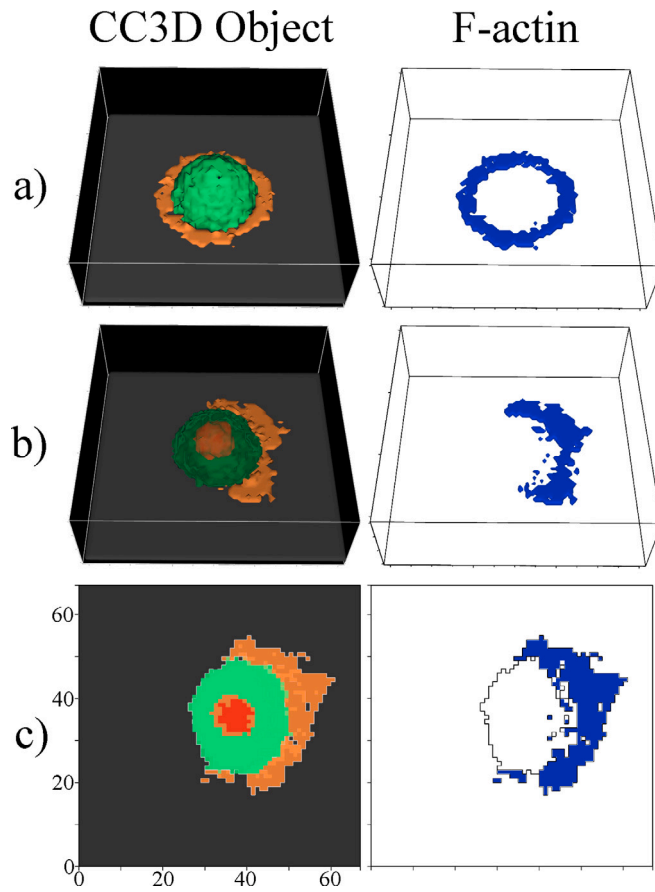


Fig. 1. Visualization of the CC3D objects composing the simulated cell and the internal field F-actin. **(a)** The cell initialization is a symmetrical semi-sphere over the flat substrate. **(b)** Cell in a polarized state. Image was edited to show the Nuc inside. **(c)** Cell in a polarized state (view of three 2D layers superposed in the xy plane). In all cases, the internal field F-actin has value 1 (blue) in the Lamellipodium voxels and 0 (white) otherwise. Note that, in the polarized state, the Nuc lags behind inside the Cyto, enabling the definition of polarization in Eq. (5), which we use to define the cell reference frame.

• External Chemical Field (NEW)

- **Properties:** concentration (continuous between 0 or 1). Constant gradient, pointing to the x direction.
- **Behaviors:** none.
- **Interactions:** is sensed by the voxels of the base of the cell, ultimately regulating where the Lamel voxels will be nucleated.
- **Dynamics:** static in time.

In CC3D, protrusion forces that arise from field gradients are modeled by an energy increment

$$\Delta E_{\text{protrusive}} = -\lambda_{F-\text{actin}} [F(\vec{i}_{\text{Medium}}) - F(\vec{j}_{\text{Lamel}})] \quad , \quad (1)$$

where $F(\vec{i}_{\text{Medium}}) - F(\vec{j}_{\text{Lamel}})$ represents a field gradient between the neighboring voxels of Medium and Lamel at sites $\vec{i}$ and $\vec{j}$, respectively. In our simulation, F is the F-actin field and $\lambda_{F-\text{actin}}$ controls the intensity of the protrusion force. This dynamics makes the cell a self-polarizing structure with a preferential migration axis that changes direction over time. Furthermore, it mimics a Local-Excitation Global-Inhibition (LEGI) dynamics [18–22]. Fig. 1 shows the cell structure and internal F-actin field in 3 different perspectives. For more details, see Refs. [3,8,14] and the supplemental materials [17].

Cell kinetics resulting from the model described in Ref. [8] agrees with experiments for cells in the absence of external fields. However, the model does not have any mechanism to promote cell response to environmental cues. Eukaryotic cell migration requires cytoskeleton organization. In several eukaryotic cell species, PI_3K , PIP_3 , Rac and Rho GTPases implement a Local Excitation Global Inhibition (LEGI) dynamics [18–22] that regulates lamellipodium polymerization and depolymerization. This way, a large and localized lamellipodium promotes its own localized growth while inhibiting growth at other locations. For more details, see Ref. [4]. In chemotaxis, the cell spatially detects the chemical gradients around the cell. This signal is biochemically transduced and perturbs the internal cell machinery, favoring a given orientation for the growth of lamellipodium, directing migration. Here, we adapt the model to simulate this three-step dynamics (sensing, reorientation, and migration [23,24]).

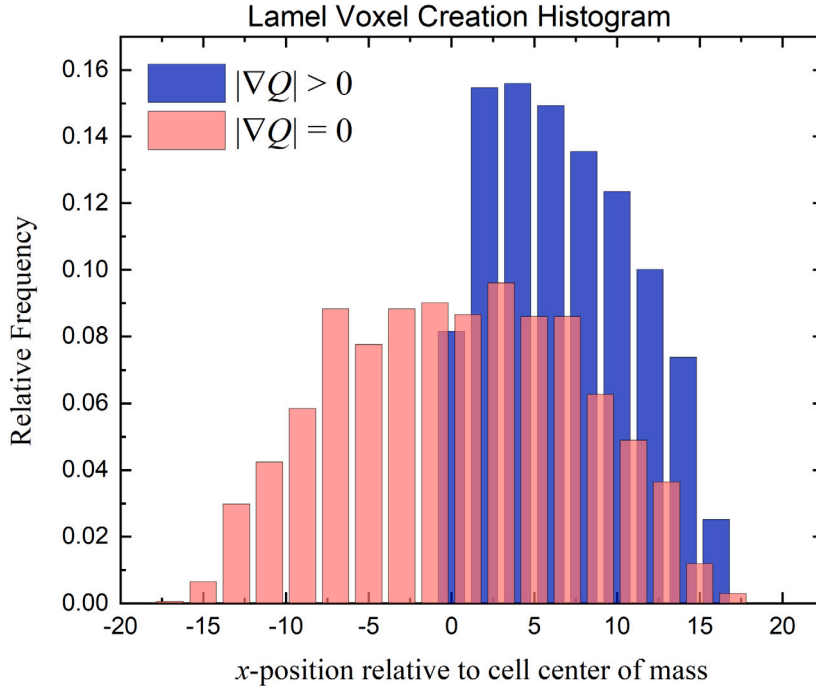


Fig. 2. Histogram of the x -position (relative to the cell's center of mass) of Lamel voxel creation in the base of the Cyto. The isotropic condition (red) makes Lamel creation random. This can be seen by setting $Q(\vec{i}) - \langle Q \rangle_{cell} = 0$ in Eq. (2). An external gradient makes Lamel creation more likely to occur where the field is more intense, reorienting the cell in the long run. This graph is obtained for a cell with $\mu = 10^6$, therefore, Lamel voxels are created only in the side of the cell with higher field intensity.

We define a linear constant external chemical field $Q(\vec{i})$, whose concentration is sensed in the cell base, i.e., all voxels of the cell (Cyto or Lamel) in the xy plane of $z = 1$. The concentration is standardized, and controls where new Lamel voxels are nucleated. In Fig. 2, we show the asymmetry of Lamel voxel creation when the external chemical field is present. The CC3D simulation is accessible on GitHub [25].

To control the production of Lamel voxels in response to chemotaxis, we developed a minimal chemical field sensing equation that controls the probability of nucleating a Lamel voxel in the Cyto's base

$$P_{convert}(\vec{i}) = \rho \left[\tanh \left(\mu \frac{Q(\vec{i}) - \langle Q \rangle_{cell}}{\sigma_{Q,cell}} \right) + 1 \right] \frac{\langle \phi_f \rangle_{n-100}^n - \phi_f^n}{|\langle \phi_f \rangle_{n-100}^n - \phi_f^n|} . \quad (2)$$

The first factor on the r.h.s. accounts for the directionality in the Lamel creation at Cyto's base. The external field $Q(\vec{i})$ is evaluated at every voxel $\vec{i}$ and is standardized. The fraction on the far right is a switch that only allows the creation of new Lamel voxels when the current Lamel volume fraction falls below an average. This equation does not participate in the calculation of the voxel copy probabilities; it controls the creation of new Lamel voxels and works independently of CPM dynamics. More details and justifications are presented in the supplemental materials [17].

Note that the chemical field sensing mechanism, given by Eq. (2), controls the nucleation of Lamel voxels, which reorients the Lamel compartment, and this is distinct from migration. After sensing and reorienting, migration is accomplished by the F-actin-promoted protrusions in the Lamel. This model achieves chemotactic response as shown in Fig. 6 trajectories in Section 3.0.5.

2.1.2. Simulation execution time

The simulation execution time, in a Potts simulation, typically scales with the number of lattice sites $N_{voxels} = L_x \times L_y \times L_z$, but can change depending on plugins and routines used. In our simulations, N_{voxels} depends on the cell size, because we adjust the lattice to the cell. One simulation with a single cell of radius 10 and 10^6 MCS, on an Intel i7-3770K processor takes approximately 4 h. 10^6 MCS of simulation time was enough to capture all expected regimes of movement — short time scale diffusive, intermediate time scale ballistic, long time scale diffusive and long time scale ballistic (if external gradient is active). Although MCS is not directly translatable to hours in real time, time can be expressed in natural units of the parameters of the system, as we show in Eq. (17).

2.2. Quantitative characterization of cell movement

A common experimental setup for studying cell migration consists in cultivating cells on flat adherent substrates in a Petri dish. For low cell density, single cell migration takes place and chemical gradients can be imposed with external control [15,16,26,27].

Cells are monitored by microscope and their positions are tracked by time-lapse imaging. Internal cellular chemical species such as ERK, F-actin, and membrane receptors can also be monitored using fluorescence techniques [28–30]. After cell trajectories are obtained, it is possible to characterize cell movement.

2.2.1. Mean square displacement — MSD

A plot of MSD versus time interval Δt shows different movement regimes indicated by its slope on log–log scale. The ballistic regimes have slope equal to 2, while the diffusive regimes have slope equal to 1. The mathematical definition of MSD is

$$\langle |\Delta \vec{r}|^2 \rangle(\Delta t) = \left\langle \frac{1}{N - \Delta t} \sum_{i=1}^{N-\Delta t} |\vec{r}(t_i + \Delta t) - \vec{r}(t_i)|^2 \right\rangle, \quad (3)$$

where the brackets represent ensemble averages, in our case, over 10 simulations. $\vec{r}(t)$ is the cell position at time t_i (subscript i indicates the discrete counting of time: $t_1, t_2, \dots$). MSD curves can be fit using models such as the Fürth equation [2], from which it is possible to calculate persistence and diffusivity. In the Supplementary Materials [17], we show a step-by-step fitting process of a MSD curve with 4 regimes of movement.

2.2.2. Mean velocity and optimal velocity

Normally, we would consider instantaneous velocity a possible characterization measurement, but cells can present diffusive behavior for short time scales [3]. Instead, we measure mean velocity using

$$\vec{V}(t, \delta) = \frac{\vec{r}(t_i + \delta) - \vec{r}(t_i)}{\delta}, \quad (4)$$

where δ is the time interval between two cell positions.

It is possible to define an optimal mean velocity $\vec{V}_{opt} = \vec{V}(t_i, \delta_{opt})$, where δ_{opt} is the time interval at which the MSD slope is the steepest (most ballistic-like). If the cell is ballistic for short time scales, $\vec{V}_{opt}$ will be taken as the limit of $\vec{V}$ for $\delta \rightarrow 0$, as normally used.

2.2.3. Polarization, polarization direction distribution, parallel and perpendicular velocity

Polarization is the preferential direction of internal cellular fibers that participate in the transport of molecules necessary to build the actin network at cell front and to retract it at cell rear. This direction correlates with cell displacement [14,31]. In our simulations, due to cell movement, Nuc tends to lag behind of the cell's geometrical center. As in Refs. [8,14], we take advantage of Nuc localization to define polarization as

$$\vec{P} = \vec{r}_{CN} - \vec{r}_N, \quad (5)$$

where $\vec{r}_N$ is Nuc's center of mass position and $\vec{r}_{CN}$ is the center of mass position of Nuc and Cyto combined. This definition leads to a vector correlated with cell displacement [14]. Experimentally, this definition requires the nucleus and cell border to be visible throughout the experiment. The distribution of chemicals within the cell is also a possible asset for determining cell polarity.

In our model, the cell can self-polarize, it does not require the external field. The spontaneous symmetry break of the polarization emerges from the protrusion forces exerted on the boundaries of the Lamel. Because the Lamel has a global volume constraint, the system is unstable and the bigger protrusion wins. It is a Local-Excitation Global-Inhibition system.

In directed migration, the histogram of polarization angle with the x -axis of the laboratory reference frame, θ , is a robust measurement of directed migration. A directional cell movement will lead to a spike in the distribution of θ at the direction of the gradient, while a cell movement in the absence of chemical gradients leads to a flat distribution of θ .

With Eqs. (4) and (5), we define parallel velocity $V_{\parallel}$ — the velocity component in the direction of polarization,

$$V_{\parallel}(t_i, \delta) = \frac{\vec{V}(t_i, \delta) \cdot \vec{P}}{|\vec{P}|}, \quad (6)$$

and $V_{\perp}$ — the velocity component orthogonal to polarization,

$$V_{\perp}(t_i, \delta) = \frac{\vec{V}(t_i, \delta) \cdot (\vec{P} \times R(\pi/2))}{|\vec{P}|}, \quad (7)$$

where $R(\pi/2)$ is a $\pi/2$ rotation matrix counterclockwise.

2.2.4. Mean velocity autocorrelation function (mVACF) and displacement autocorrelation function

An important measurement that considers cell mean velocity is the Mean Velocity Autocorrelation Function — mVACF. mVACF can be used to calculate the characteristic memory time and to determine underlying dynamics [32]. It is defined as

$$\text{mVACF} = \left\langle \frac{1}{N - \Delta t - \delta} \sum_{i=1}^{N-\Delta t-\delta} \vec{V}(t_i, \delta) \cdot \vec{V}(t_i + \Delta t, \delta) \right\rangle, \quad (8)$$

where $\vec{V}(t_i, \delta)$ follows the definition of mean velocity in Eq. (4). mVACF remains as a function of Δt — the time interval between two velocity measurements — and δ — the time interval used to measure mean velocity in Eq. (4). Note that $\Delta t > \delta$ so that two successive

mean velocity measurements do not overlap in time, which would introduce an artificial correlation [3]. The summation is taken over all N points of the trajectory, except when $t_i > N - \Delta t - \delta$, for which the next mean velocity measurement would fall outside the trajectory.

Since cells are often polarized structures, another autocorrelation to consider is of cell displacement in the directions parallel and perpendicular to polarization. This metric extracts further information about the system's underlying dynamics. Using brackets $\langle \rangle$ to denote ensemble averages, we define the displacement autocorrelation function as

$$C_{rr} = \langle (\Delta \vec{r}_t \cdot \vec{n}_t)(\Delta \vec{r}_{t+\Delta t} \cdot \vec{n}_{t+\Delta t}) \rangle \quad (9)$$

for the direction parallel to the polarization $\vec{n}$, defined in Eq. (5) (see following subsection). For perpendicular direction, C_{rr} is analogous, but considers $\vec{n}$ rotated 90° in the xy plane.

2.2.5. Drift speed

We define drift speed as the average (over time and over realizations) of the mean velocity projected on the polarization vector for $\delta \rightarrow 0$:

$$V_d = \left\langle \lim_{\delta \rightarrow 0} \frac{1}{N - \delta} \sum_{i=1}^{N-\delta} V_{\parallel}(t_i, \delta) \right\rangle. \quad (10)$$

$V_{\parallel}(t_i, \delta)$ is the parallel velocity defined in Eq. (6). The resulting scalar V_d measures how fast the cell moves in the polarization direction in average, which may converge for small time intervals $\delta \rightarrow 0$ even if the movement is diffusive for short time intervals. That could happen when the diffusive contribution for displacement has zero average. If the cell displaces $\Delta \vec{r}_d$ in the direction of polarization due to the drift plus a random diffusion $\Delta \vec{\xi}$ during a time interval equal to δ . Then,

$$\left\langle \vec{V}(t_i, \delta) \cdot \frac{\vec{n}_i}{|\vec{n}_i|} \right\rangle = \left\langle \left(\frac{\Delta \vec{r}_d}{\delta} + \frac{\Delta \vec{\xi}}{\delta} \right) \cdot \frac{\vec{n}_i}{|\vec{n}_i|} \right\rangle = \left\langle \frac{\Delta \vec{r}_d}{\delta} \cdot \frac{\vec{n}_i}{|\vec{n}_i|} \right\rangle, \quad (11)$$

since the noise term average is zero. The ensemble average allows the convergence of $\left\langle \vec{V}(t_i, \delta) \cdot \frac{\vec{n}_i}{|\vec{n}_i|} \right\rangle$ even though $\frac{\Delta \vec{\xi}}{\delta}$ diverges for $\delta \rightarrow 0$. Consequently V_d is well defined.

If the cell responds to an external chemical gradient but no significant change in drift speed is observed, it must be the case that cell's polarization is aligned to the gradient. That is why analyzing both polarization angle distribution (previous item) and drift speed is important to fully understand how cells respond to stimuli.

2.2.6. Terminal speed and the chemotactic efficiency

When cells respond to external chemical gradients, a non zero terminal speed appears, causing a MSD curve with slope equals to 2 for large Δt . F. Peruani and L. G. Morelli [33] analytically demonstrated that an orientation directionality only (without changes in speed behavior) causes an extra ballistic term for long time intervals in the MSD. Therefore, we approximate the MSD in this regime to

$$\langle |\Delta \vec{r}|^2 \rangle \sim B \Delta t^2, \quad (12)$$

where $V_T = \sqrt{B}$ is the terminal speed of the cell (more details in the results section). We define the ratio between terminal speed and drift speed (defined in Eq. (10))

$$\varepsilon = \frac{V_T}{V_d}, \quad (13)$$

as a measurement of chemotactic efficiency. The idea behind this metric is that a cell perfectly aligned to the chemical gradient will convert all its drift speed into terminal speed, whereas a less aligned cell will certainly have a terminal speed smaller than its drift speed. Provided that the cell has a net velocity (drift speed) in the direction of polarization, this metric can be applied to any taxis mechanism, since every directed migration will appear as a long-term ballistic MSD curve.

2.2.7. Cell displacement distributions and cell reference frame

The investigation of cell movement with respect to the laboratory reference frame and to the cell reference frame requires a detailed set of measurements. Here we present the distribution of the cell displacement for a given time interval Δt as our final way to quantify and visualize the cell behavior in the 2D plane. To perform this metric, we (1) take the cell's trajectory with the polarization vector in each time step; (2) choose a set of time intervals Δt to analyze; (3) calculate the tuples $(\Delta \vec{r}_x, \Delta \vec{r}_y)$ and $(\Delta \vec{r}_{\perp}, \Delta \vec{r}_{\parallel})$ for all time points and trajectories and every chosen Δt , where the last tuple is the parallel and perpendicular component of the displacement relative to the polarization vector measured in the beginning of each step Δt ; (4) perform the 2D frequency counts of the tuples of $\Delta \vec{r}$ for the two coordinate systems separately; and (5) plot resulting 2D histograms.

When the displacement is projected onto the polarization direction, a new reference frame emerges, where the x -coordinate is the displacement component in the direction of polarization and the y -coordinate is the displacement component in the perpendicular direction of polarization. This strange non-inertial reference frame allows us to further comprehend cell migration dynamics, as it is inextricably linked to polarization. As we show in the results section, the migration behavior of our cells is not affected by the external chemical gradient when analyzed in the cell reference frame. We suspect this might also be true for experimental cells. However, polarization measurement is required to perform our tests. Current literature on single cell chemotaxis (or any other form of directed migration) does not provide datasets where both cell position and cell polarization are measured throughout the entire trajectory.

3. Results

3.0.1. Varying cell radius, Lamel volume and protrusion strength modify cell migration

To characterize cell movement, we measured the mean velocity components relative to polarization and we compare the average of mean velocity with the average of absolute mean velocity for all available time intervals Δt .

We varied three parameters, as shown in Fig. 3. First, we observed an increase in drift speed with $\lambda_{F-actin}$ — the protrusion coefficient. This happens because $\lambda_{F-actin}$ regulates the likelihood of Lamel voxels being copied over Medium voxels. If $\lambda_{F-actin}$ is too high ($\lambda_{F-actin} > 175$), unwanted effects take place such as the breaking between Lamel and Cyto (also observed in [8]). Second, we observe a decrease in drift speed for ϕ_f too large or too small. For too large ϕ_f , Lamel organizes as a ring around the cell, disfavoring polarization. In the opposite case, if ϕ_f is too small, there will not be enough Lamel to protrude, reducing drift speed. Lastly, cell radius changes the relative effect of the membrane fluctuations and Lamel's size. For larger cell radii of $R = 15$ and $R = 20$, the area covered by Lamel is comparatively larger than $R = 10$ because ϕ_f grows with R^3 while Lamel is flat, and the average membrane fluctuation will have a reduced comparative effect over the larger cells because it only acts on the interface between two CPM objects. Therefore, consistent scaling of the system with the cell by radius is non-trivial. For this reason, next results will be restricted to $R = 10$.

3.0.2. Short time diffusive behavior is isotropic

The simulated cell has a polarization direction defined by Eq. (5), with distinct dynamics for parallel and perpendicular directions with respect to it. By projecting the mean velocity of the cell in both perpendicular ($V_\perp$) and parallel ($V_\parallel$) directions to the cell polarization at the beginning of the time interval, we find that diffusive noise is present in both directions, shown by the divergence in the moduli of both velocity components. Fig. 3 shows $\langle |V_\perp| \rangle$ and $\langle |V_\parallel| \rangle$ as functions of the time interval δ used to calculate mean velocities for different simulation parameters. We observe that both quantities diverge as $\Delta t \rightarrow 0$, implying that instantaneous velocity cannot be defined in any direction. A similar figure, but with double logarithmic scale, is shown in the Supplemental Materials [17], where the divergence behavior of $\langle |V_\parallel| \rangle$ and $\langle |V_\perp| \rangle$ is clearer.

3.0.3. The modulus of finite drift speed is not significantly affected by the external field

To investigate the effect of an external chemical field $Q(\vec{r})$ (constant in time, linear in x -direction), we set a saturation value $\mu = 10^6$ for Eq. (2). We measure the mean velocity as in Section 3.0.2, the distribution of polarization angle θ , and we compare the sensitive cell $\mu = 10^6$ subjected to the gradient to the isotropic case (no external chemical gradients).

We show in Fig. 4 that the drift speed (defined in Eq. (10)) does not change significantly when the gradient is active $\vec{\nabla}Q > 0$ (lines) compared to the isotropic case $\vec{\nabla}Q = 0$ (dots). The same applies to the average of the absolute mean velocities $\langle |V_\parallel| \rangle$ and $\langle |V_\perp| \rangle$ for all studied ϕ_f and $\lambda_{F-actin}$. Based on the observed data, we concluded that the changes in the parallel velocity is not the primary cause of chemotactic response. The next step is to analyze polarization, as it could promote chemotaxis if a reorientation dynamics exists.

3.0.4. Chemotaxis does not act as an external force

To test the role of cell's orientation, we measured θ — the direction of cell's polarization (see Eq. (5)) — throughout the simulations and made a histogram to check cell alignment with the field gradient. Fig. 5 shows that the polarization angle θ relative to the x -direction has a flat distribution in the absence of external field, in contrast to a peaked distribution in the direction of the gradient when external field is present. The results in Figs. 4 and 5 suggest that the cell's migration dynamics in respect to polarization is preserved during chemotactic response, a topic we will further explore later. For now, we can conclude that our chemotaxis mechanism acts as guidance for cell polarity, not as an external force acting over cell's center of mass.

3.0.5. Trajectories and MSD show long time ballistic regime in the presence of an external field

To numerically characterize cell kinetics, we employ MSD — Mean Square Displacement measurements and fit procedures based on mathematical models for cell kinetics. We also use MSD fit results to convert our data to natural units, allowing comparison between different computational cell models and experiments.

MSD curves from experiments show three kinetic regimes for cell migration when external field gradient $|\vec{\nabla}Q|$ is zero: diffusive at short time intervals, ballistic-like at intermediary time intervals, and diffusive at long time intervals [3,8]. The present simulations show the same behavior, in Fig. 6 (red dots). On the other hand, a fourth ballistic regime appears for time intervals long enough if an external field gradient $|\vec{\nabla}Q| > 0$ is present. We fit the MSD curves using the equation

$$\langle |\Delta \vec{r}|^2 \rangle = 2D(\Delta t - P(1 - e^{-\Delta t/P})) + \frac{2DS}{1-S}\Delta t + B\Delta t^2, \quad (14)$$

where the first term is the F urth equation — solution of the Langevin equation — with diffusion constant D and persistent time P . The second term linear in Δt accounts for the first diffusive regime, with diffusion constant $\frac{DS}{1-S}$, as proposed in Ref. [3]. The third term accounts for the external gradient response, where $V_T = \sqrt{B}$ is the terminal speed, as defined in Eq. (10). In Supplementary Materials [17], we show a step-by-step fitting process of this MSD curve, and Tables S1 and S2 show all numerical results.

Taxis time t_{axis} is defined as the time interval that separates the third regime (long term diffusive) and the fourth ballistic regime associated to chemotactic response; we calculate it using

$$t_{axis} = \frac{2D}{B(1-S)}. \quad (15)$$

We end up with four characteristic time intervals:

Mean Velocity Analysis

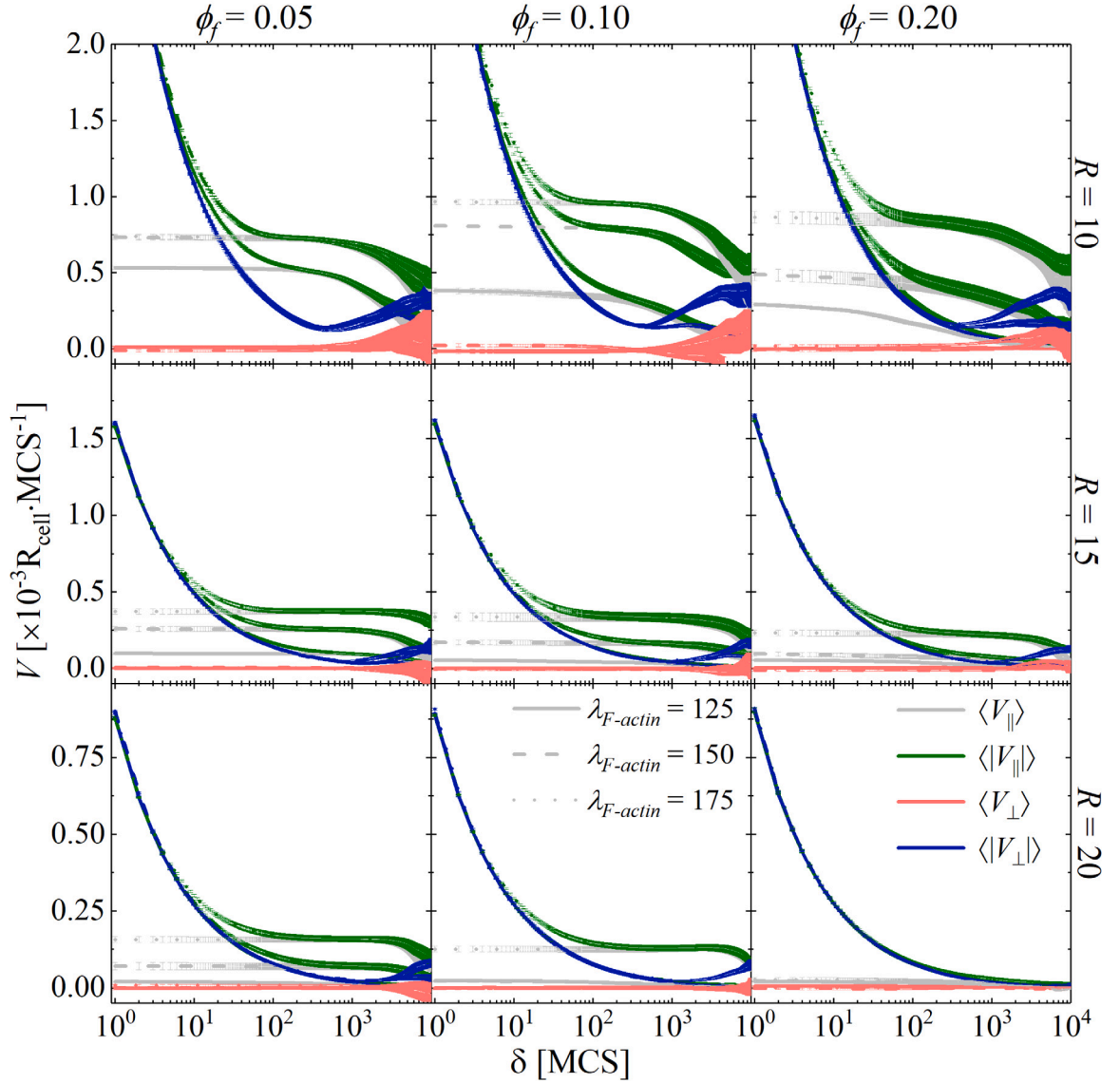


Fig. 3. Four different measurements as functions of δ , the time interval used to calculate the mean velocity: (1) $\langle V_{\parallel} \rangle$ is the average of the parallel component of the mean velocity with respect to the polarization vector (gray); (2) $\langle |V_{\parallel}| \rangle$ is the same, but the absolute value is taken before averaging (green); (3) $\langle V_{\perp} \rangle$ is the average of the perpendicular component of the mean velocity with respect to the polarization vector (red); (4) $\langle |V_{\perp}| \rangle$ is the same, but the absolute value of velocity is taken (blue). We measured mean velocity for three different values of relative Lamel volume ϕ_f (graph columns), Lamel protrusion coefficient $\lambda_{F-actin}$ (different line styles), and different cell radius (graph rows). Divergence in $\langle |V_{\parallel}| \rangle$ and $\langle |V_{\perp}| \rangle$ as $\delta \rightarrow 0$ demonstrates the isotropic diffusive noise, unlike $\langle V_{\parallel} \rangle$ and $\langle V_{\perp} \rangle$, which converge to V_d and zero, respectively. Following Eq. (10), we extract drift speed V_d from the graphs by taking the $\langle V_{\parallel} \rangle$ value for the smallest δ , since it converges. A similar figure, but with double logarithmic scale, is shown in the Supplemental Materials [17], where the divergence behavior of $\langle |V_{\parallel}| \rangle$ and $\langle |V_{\perp}| \rangle$ is clearer.

1. $\Delta t = SP$: the time interval that separates the first diffusive regime and the intermediate ballistic.
2. $\Delta t = \delta_{opt}$: the time interval where the intermediate ballistic regime has the highest slope, we call it “optimal delta”.
3. $\Delta t = P$: the persistence time, which separates the intermediate ballistic regime and the long-term diffusive regime.
4. $\Delta t = t_{axis}$: the taxis time, which separates the long-term diffusive regime and the long term ballistic regime associated with the chemotaxis response.

Mean Velocity Analysis - Gradient (R=10)

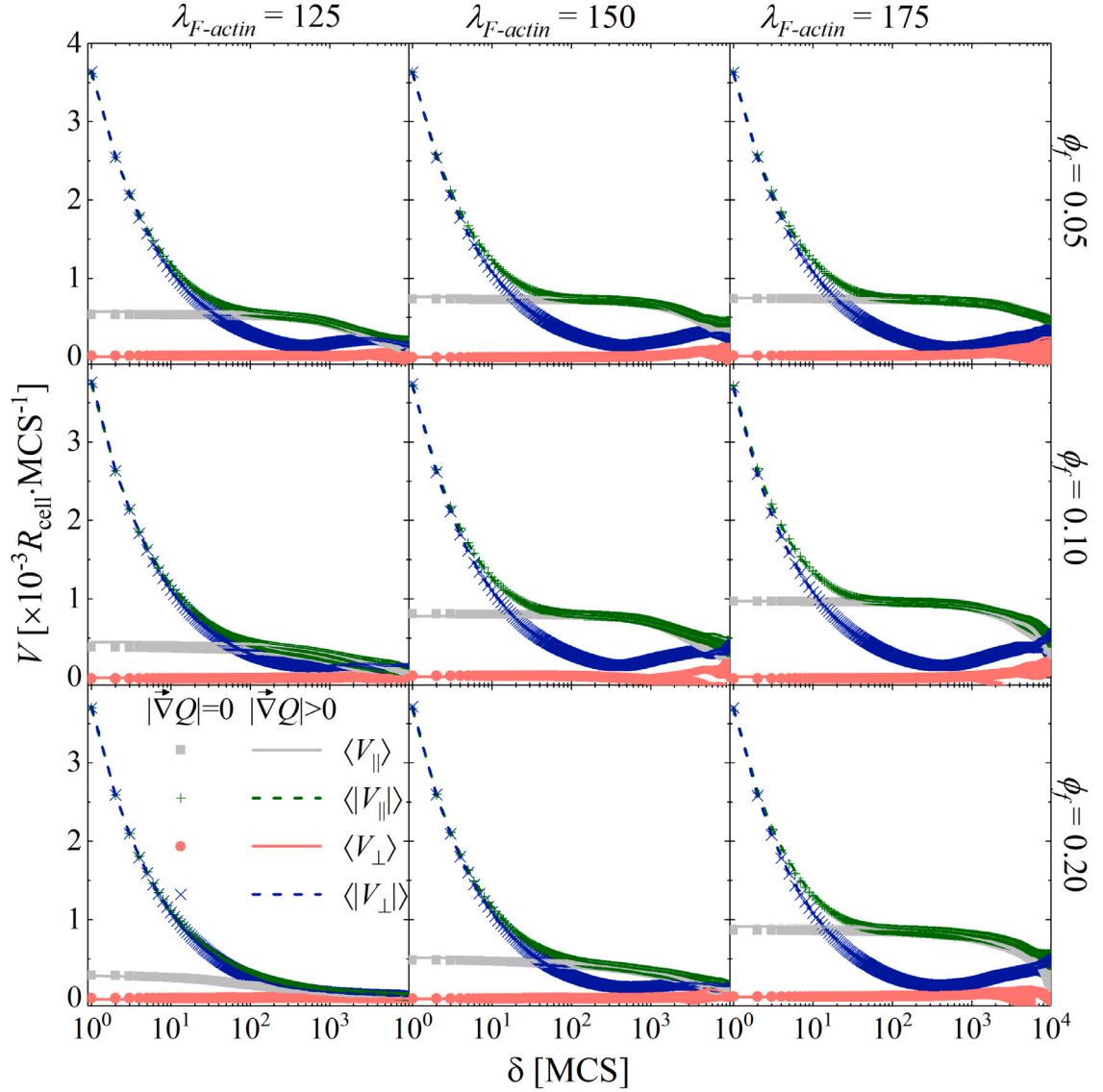


Fig. 4. Average mean velocity in parallel and perpendicular direction to polarization (represented with different colors) depending on Lamel protrusion coefficient $\lambda_{F-actin}$ (graph columns) and on the relative Lamel volume ϕ_f (graph rows), for two cases: (1) environment is isotropic $\vec{\nabla}Q = 0$ (points), and (2) the cell is exposed to an external chemical gradient $\vec{\nabla}Q > 0$ (lines). The parallel velocity $\langle V_{\parallel} \rangle$ (gray) shows little difference between the two cases for every parameter set, suggesting that the external gradient does not increase cell's drift speed. $\langle V_{\parallel} \rangle$ (green) and $\langle V_{\perp} \rangle$ (blue) also do not show significant differences between the two cases, which means the external gradient did not affect the diffusive dynamics as well.

If the MSD presents these 4 regimes in the correct order, we expect that

$$SP < \delta_{opt} < P < t_{axis} \quad . \quad (16)$$

The MSD curves in Figs. 6 and 7 are presented in natural units of

$$\langle \Delta \vec{r}^2 \rangle = \langle \Delta \vec{r}^2 \rangle \frac{1-S}{2DP} \quad , \quad \text{and} \quad \Delta \tau = \Delta t / P \quad , \quad (17)$$

which collapse all curves into one if $S = 0$ (Langevin limit) and $|\vec{\nabla}Q| = 0$ (isotropy). Since each cell has a different S , curves separate for small time intervals, but remain united from $\Delta \tau = 1$ forward. If cells have a terminal speed due to external field, then each cell will have a different value for B as well, making curves separate after $\Delta \tau = 1$. In Fig. 7 we show the curves collapsing for both cases: isotropic ($|\vec{\nabla}Q| = 0$) and anisotropic ($|\vec{\nabla}Q| > 0$).

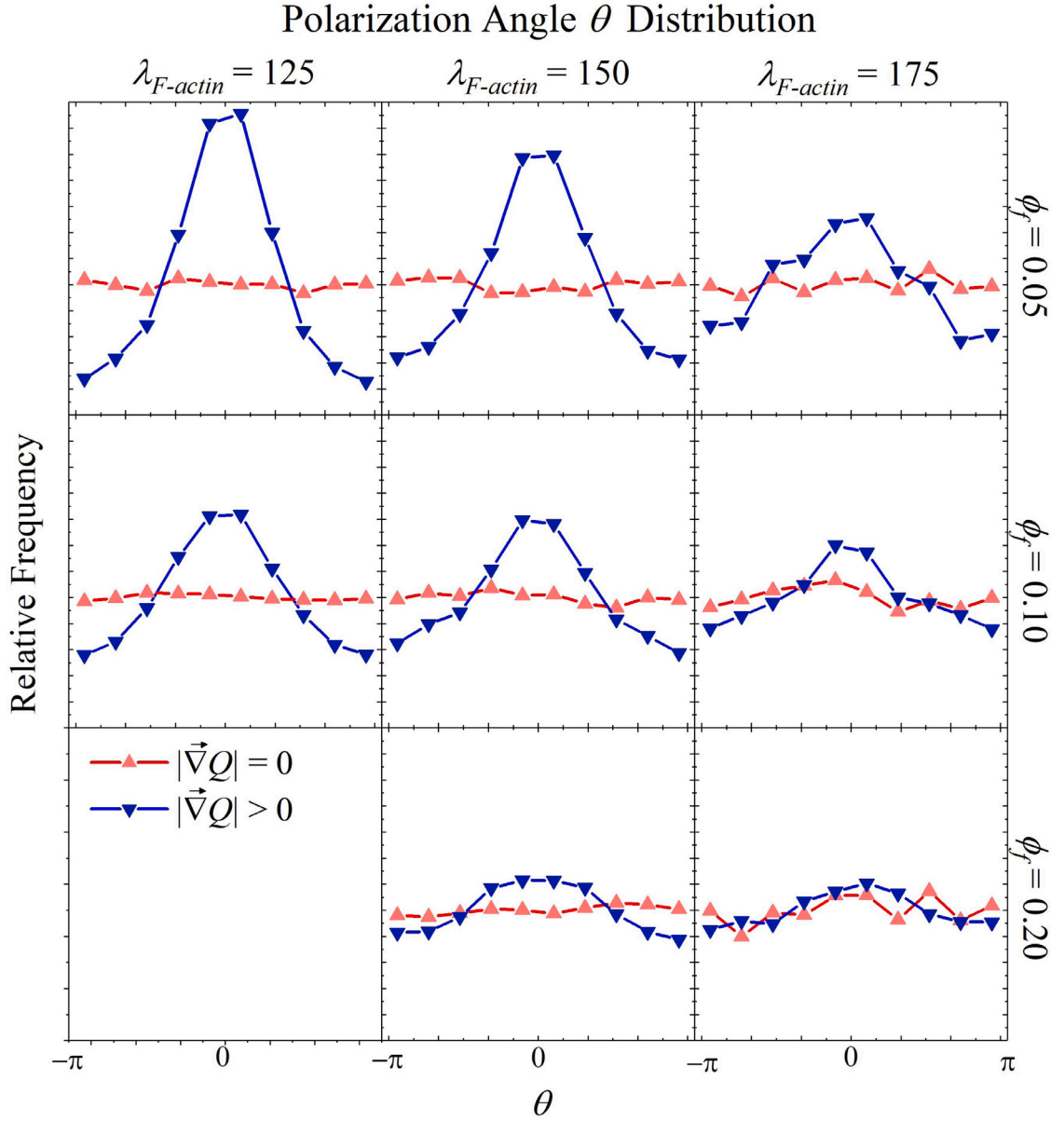


Fig. 5. Angle θ of the polarization vector with respect to the x -axis for different values of $\lambda_{F-actin}$ (graph rows) and ϕ_f (graph columns). Since the external field is linear in the x -coordinate, θ is also the angle between the polarization vector the chemical gradient. For $|\vec{\nabla}Q| > 0$ (blue curves and points), θ distribution is narrow around $\theta = 0$, while for $|\vec{\nabla}Q| = 0$ (red curves and points), θ distribution is flat. This figure indicates that the chemical gradient acts reorienting cell polarization.

In the absence of an external field, the long-term diffusive regime is due to the loss of polarization memory. In the presence of an external field, the chemical gradient becomes a preferential direction for polarization to align. Since the cell moves preferentially in the direction of polarization, a ballistic regime appears for very long time intervals, that is, $MSD \propto \Delta\tau^2$. Fig. 6 also shows some typical trajectories for each set of parameters, in the presence (blue) and absence (red) of an external chemical field to illustrate the change in behavior due to the action of the external field.

3.0.6. The chemotactic efficiency is a robust metric and depends on the cell parameters

Here we apply our chemotactic efficiency ε metric, defined in Eq. (13), to our simulated cells. In Fig. 8, we show the relation between ε and the parameters studied: protrusion coefficient $\lambda_{F-actin}$ and relative Lamel volume fraction ϕ_f .

The reason why ε decreases with ϕ_f is related to the behavior of Lamel as a function of its size. As ϕ_f increases, Lamel spreads more around Cyto, hindering polarization and reducing the relative effect of newly created Lamel voxels.

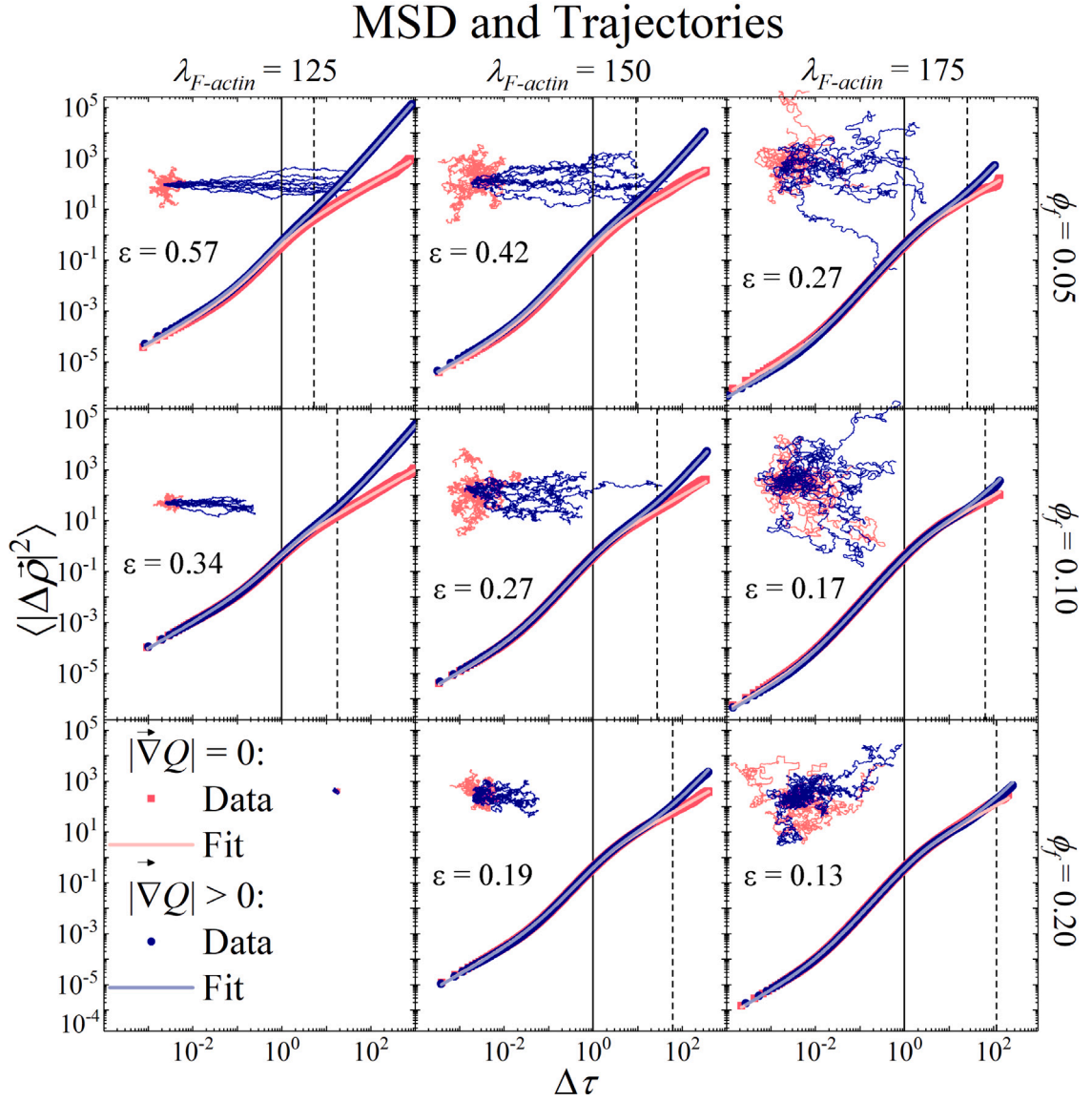


Fig. 6. Mean Square Displacement simulation data, fit curves and corresponding trajectories. Natural units for distance and time are such that $\Delta \tau = \Delta t/P$ and $\langle \Delta \rho^2 \rangle = \langle \Delta r^2 \rangle (1 - S)/2DP$. The solid vertical lines divide the persistent regime and the long term diffusive regime ($\Delta \tau = 1$). The dashed vertical lines divide the long term diffusive regime and the long term ballistic regime (only valid when $|\nabla Q| > 0$). Dashed lines occur at $\Delta \tau = t_{axis}/P$. Notice the trajectories for the set $\lambda_{F-actin} = 125$ and $\phi_f = 0.2$, they are the small dot beside the legend. We do not show the MSD for this specific set because it is impossible to find its natural units using Eq. (3) fit. We also show the chemotactic efficiency ϵ results for each case below the trajectories. In the Supplementary Materials [17], Table S1 shows all results for red points (no chemotaxis), and Table S2 shows all results for blue points (with chemotaxis). We also show each regime in isolation, with larger and cleaner plots, and a full step-by-step fitting process of an MSD curve with 4 regimes.

$\lambda_{F-actin}$ regulates the protrusive force of the F-actin gradient in the interface between Lamel and Medium, as given by Eq. (1). Greater $\lambda_{F-actin}$ means more copies of Lamel voxels on the surrounding Medium in the $z = 1$ plane. As a consequence, a newly created Lamel voxel back in Cyto will likely disappear (replaced by a Cyto voxel again) giving room to another Lamel copy towards Medium due to protrusion strength. In other words, a higher $\lambda_{F-actin}$ makes Lamel a more stable structure wherever it points to. Therefore, the Lamel creation mechanism has lower effect over strong and stable Lamel. In other words, our model presents a trade-off between mobility in the absence of external gradients and chemotactic efficiency. The results of this metric are congruent with Fig. 5, and also agree with the shape of the trajectories in Fig. 6. Therefore, this metric is robust and can provide meaningful information about directed migration.

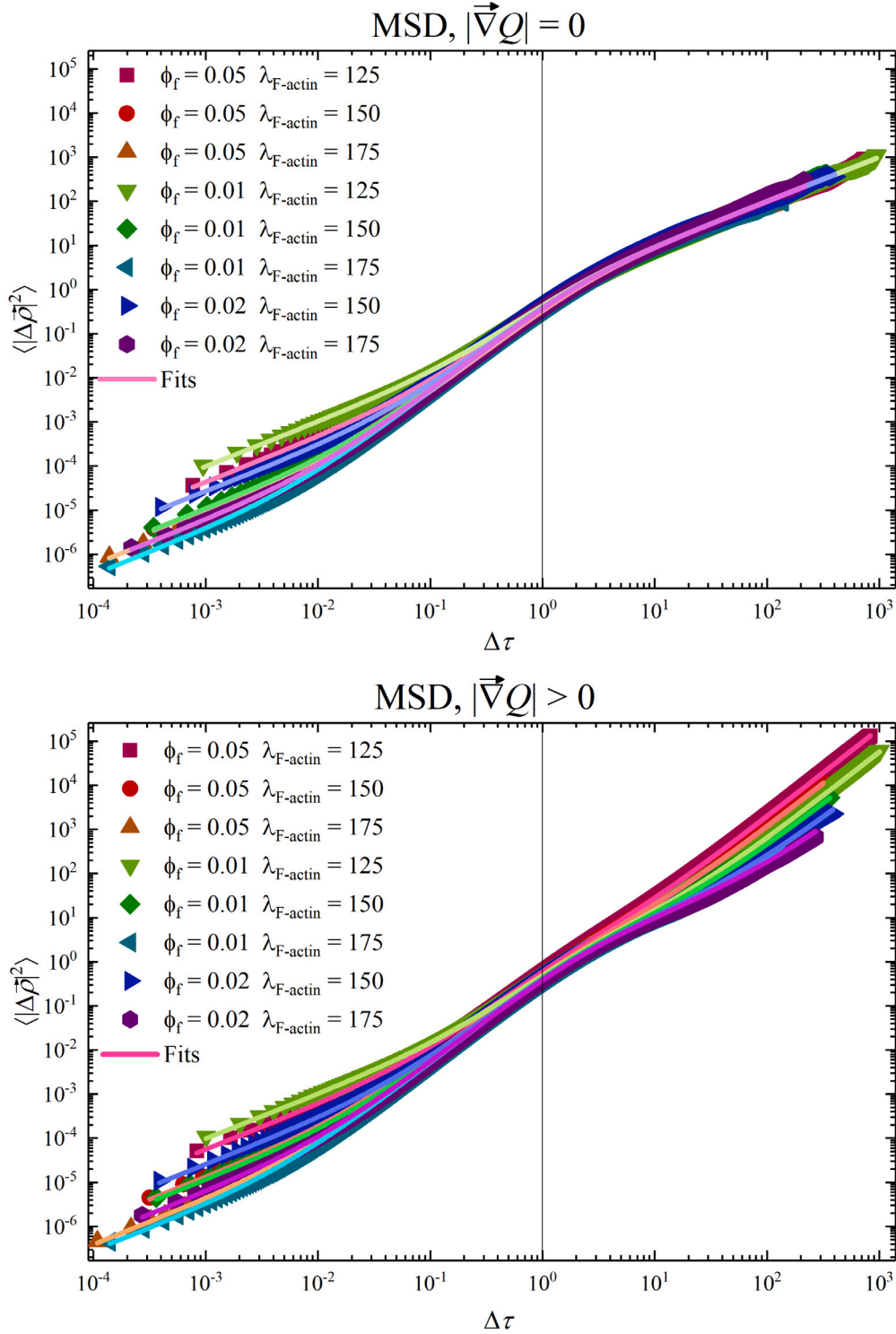


Fig. 7. All MSD curves presented together in presence (upper panel) and absence (lower panel) of external field. Natural units lead to the collapse of the curves at $\Delta\tau = 1$ (both cases) and at $\Delta\tau > 1$ (only in the isotropic case). Different values of S and B separate the curves at the respective regimes.

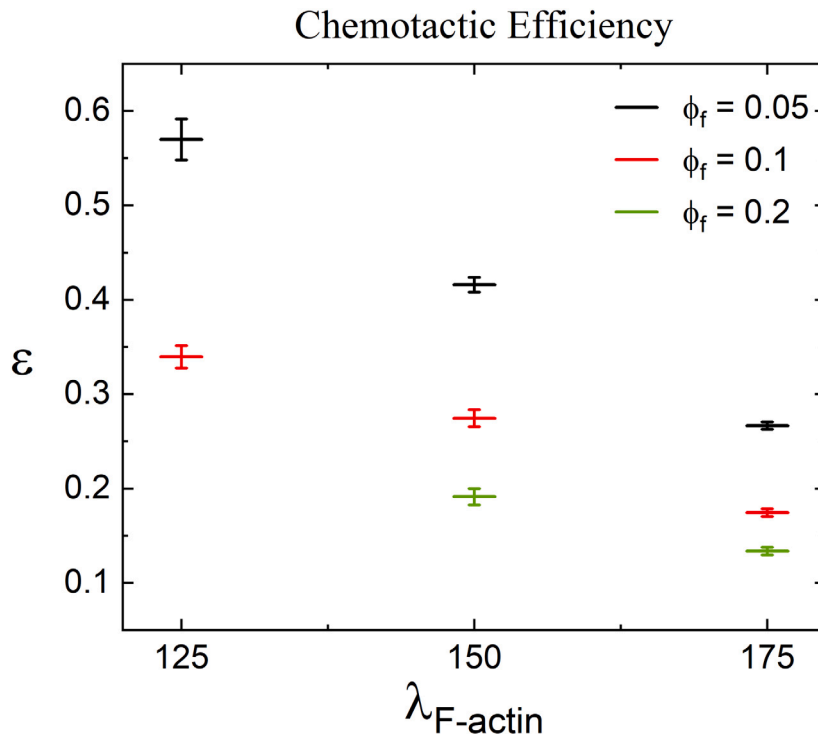


Fig. 8. Relation between chemotactic efficiency ϵ and parameters $\lambda_{F-actin}$ and ϕ_f . Simulation set with $\lambda_{F-actin} = 125$ and $\phi_f = 0.05$ does not present enough movement, so the metrics do not apply. We ran 10 simulations to generate each point.

3.0.7. mVACF shows a remaining correlation for long time intervals in the presence of external field, and displacement autocorrelation is negative for small time intervals

As we show in Fig. 9, mVACF provides further details on the loss of migration memory over time. Starting from $\Delta\tau = \delta^*$, as Δt increases, mVACF increases, reaching a maximum value, followed by an exponential decay towards an asymptotic value for large $\Delta\tau$. All cases where $|\vec{\nabla}Q| = 0$ have a zero asymptotic value, which means there is no remaining memory. However, for $|\vec{\nabla}Q| > 0$, the asymptotic value is greater than zero and depends on cell parameters. Positive mVACF for large $\Delta\tau$ indicates directed migration. We show that the asymptotic value coincides with the squared terminal speed V_T^2 . mVACF can be calculated using different δ (time interval used to measure mean velocity) and gives similar results if δ is not too small.

The behavior of mVACF for small $\Delta\tau$ implies a negative autocorrelation, hence an additional dynamical term for cell velocity. This has been observed before in [3,8,34–36]. Thomas 2019 explained the loss in measurement precision due to the higher weight of diffusive terms compared to drift terms when $\delta \rightarrow 0$. However, this effect would also rule out the possibility of measuring drift speed for small values of δ , as shown in Figs. 3 and 4.

Our hypothesis to explain this effect is that the cell dynamics for short time intervals yields a negative displacement autocorrelation function. To test this hypothesis, we measured $C_{rr}(\Delta t)$ for both parallel and perpendicular directions to polarization. Fig. 10 shows what we expected; the correlation is negative for low values of Δt . We lack a mechanistic explanation to this effect, and exploring it in depth is not within the scope of this work. Informal conversation with fellow researchers raised the hypothesis that a tendency of CPM to undo pixel copies could justify the observed effects.

3.0.8. Cell displacement probability distribution are different in the laboratory and cell reference frames

In Section 3.0.2 we inferred that our cell's dynamics is preserved during chemotactic response. To test this hypothesis, we calculate cell displacement components in two reference frames: (1) cell reference frame defined by polarization and (2) laboratory reference frame, defined by the lattice coordinates.

We chose the simulation with parameters $\phi_f = 0.1$ and $\lambda_{F-actin} = 150$ to show the displacement distributions in Fig. 11, considering four different time intervals Δt : $0.01P$ (fast diffusive), $0.1P$ (ballistic-like), P (ballistic-like), and $10P$ (slow diffusive or ballistic depending on $\vec{\nabla}Q$ conditions). We also considered two reference frames: laboratory and cell reference frames, with and without the external field. The external field gradient orientation is aligned to the x -axis of the laboratory reference frame. In the cell reference frame, the x -axis is the cell's polarization direction at the beginning of the step.

In the cell reference frame, displacement distributions change depending on time scale Δt . For small Δt , distribution is Gaussian-like and centered at a positive value of the x -axis, indicating a movement with diffusion and a drift aligned to polarization. For Δt close to persistence time, asymmetry increases, showing the correlation between polarity and displacement in these time scales.

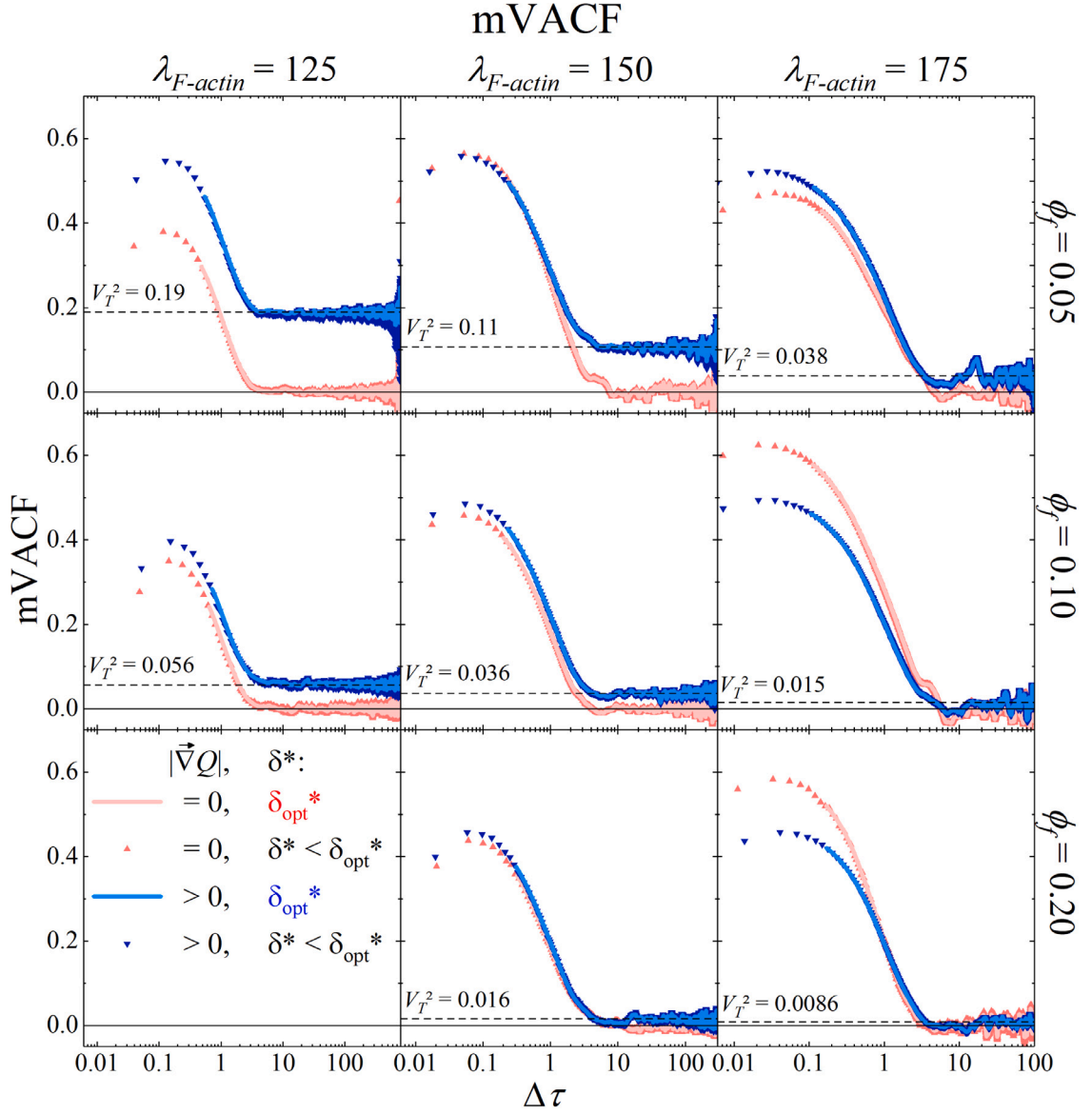


Fig. 9. Mean Velocity Autocorrelation Function. Symbols stand for mVACF using $\delta^* = 50/P$. Straight lines represent mVACF using δ_{opt}^* (in natural units) where $\Delta\tau = \delta_{opt}^*$ corresponds to time interval that yields the maximum slope of the MSD in the persistent regime. We remark that $\delta^* = 50/P < \delta_{opt}^*$. For $\Delta\tau > 0$, the $\delta^* = 50/P$ curves (dark hue symbols) show that mVACF has a maximum. All straight lines (light hues) start after the peak showed by the symbols. As $\Delta\tau \rightarrow \infty$, mVACF goes to zero in the absence of chemical gradient ($|\vec{\nabla}Q| = 0$), or to a finite value in the presence of chemical gradient ($|\vec{\nabla}Q| > 0$). In fact, the terminal speed squared V_T^2 obtained from the MSD fit coincide with the mVACF's rest value for large $\Delta\tau$ (in natural units), evidence of the quality of our fit.

For time scales at the order of $10P$, displacement distribution becomes Gaussian-like again. When comparing cases with $|\vec{\nabla}Q| = 0$ and $|\vec{\nabla}Q| > 0$, we do not see significant differences, indicating that migration dynamics with respect to polarization stays preserved regardless of the external gradient.

In the laboratory reference frame, displacement distributions also depend on time scale Δt . For short Δt , distributions are Gaussian-like, indicating again the diffusive noise. For Δt close to persistence time, we found ring-like distributions, indicating that longer steps are more likely to occur. For time scales about $10P$, distributions become Gaussian-like again. Unlike the cell reference frame, the laboratory frame shows clear differences between the $|\vec{\nabla}Q| = 0$ and $|\vec{\nabla}Q| > 0$ cases. $|\vec{\nabla}Q| = 0$ distributions are symmetric, while $|\vec{\nabla}Q| > 0$ distributions are asymmetric due to the directionality of polarization angle θ .

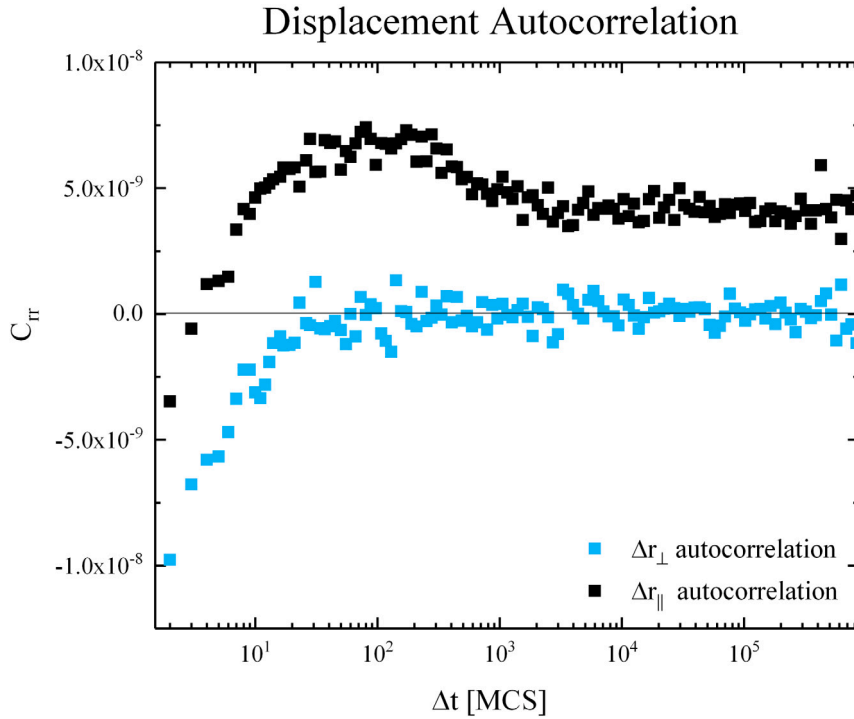


Fig. 10. Displacement Autocorrelation Function versus the time interval Δt for parameters $\phi_F = 0.1$, $\lambda_{F-actin} = 125$, without external gradient. The displacement parallel to polarization ($\Delta r_{\parallel}$) shows a positive correlation for large Δt , which confirms the presence of a drift in this direction. The displacement perpendicular to polarization ($\Delta r_{\perp}$) has zero correlation for large Δt , showing zero drift in this direction. Both displacements go to negative correlation as $\Delta t \rightarrow 0$, but $\Delta r_{\perp}$ falls faster.

4. Conclusion and discussion

4.1. Overview

We presented a model that reproduces chemotaxis behavior in a linear, constant chemical field. The cell responds to the gradient by reorienting polarization rather than being pushed by it. Directional sensing, polarity reorientation and migration are separated and simultaneous processes.

With this model, we showed it is possible to separate the cell dynamics in a new coordinate system defined by the cell's polarization. This method allowed us to identify an isotropic diffusive behavior in cell velocity. Together with this diffusive noise, we found that the cell has a net velocity only in the polarization direction, which we called drift speed.

When our simulated cell was subjected to an external chemical gradient, we found no significant change in drift speed. Then we calculated the distribution of the polarization direction, which shows a peak in the direction of the gradient. Hence, our cells respond to gradients via polarization reorientation.

The MSD showed a long-term ballistic regime for directed migration, and all curves showed the same diffusive behavior for short time scales (not following Langevin models). Nonetheless, we fit all MSD curves with a modified version of the Fürth equation and found 4 distinct time intervals of interest: (1) SP : the time interval that separates the first diffusive regime from the intermediate ballistic regime, (2) δ_{opt} : the time interval where the intermediate ballistic regime has the highest slope, (3) P : the persistence time, and (4) t_{axis} : the taxis time, which separates the long-term diffusive regime and the long term ballistic regime associated with the chemotaxis response. S and P result from the MSD fits. We also extracted V_T (terminal speed) from the MSD fits, which we then used to calculate chemotactic efficiency and taxis time t_{axis} .

We defined a robust metric of chemotactic efficiency that can be applied to any cell that has a net speed in the direction of polarization (drift speed) and responds to the chemical gradient with chemotaxis, regardless of the response mechanism. We detected a trade-off between cell polarization stability and its ability to respond to external chemical gradients.

The mVACF showed a positive correlation for long time intervals in the presence of the external chemical gradient, which is equal to V_T^2 (the terminal speed squared). It also showed that the correlation is low at short-time intervals, indicating a negative autocorrelation in the dynamics of cell displacement.

Finally, we measured the displacement distribution for different time intervals and used two reference frames of interest: the cell reference frame, defined by its polarization, and the laboratory reference frame, defined by the lattice, yielding different outcomes. The displacement distributions showed how the migration dynamics with respect to polarization remained unchanged regardless of the external gradient.

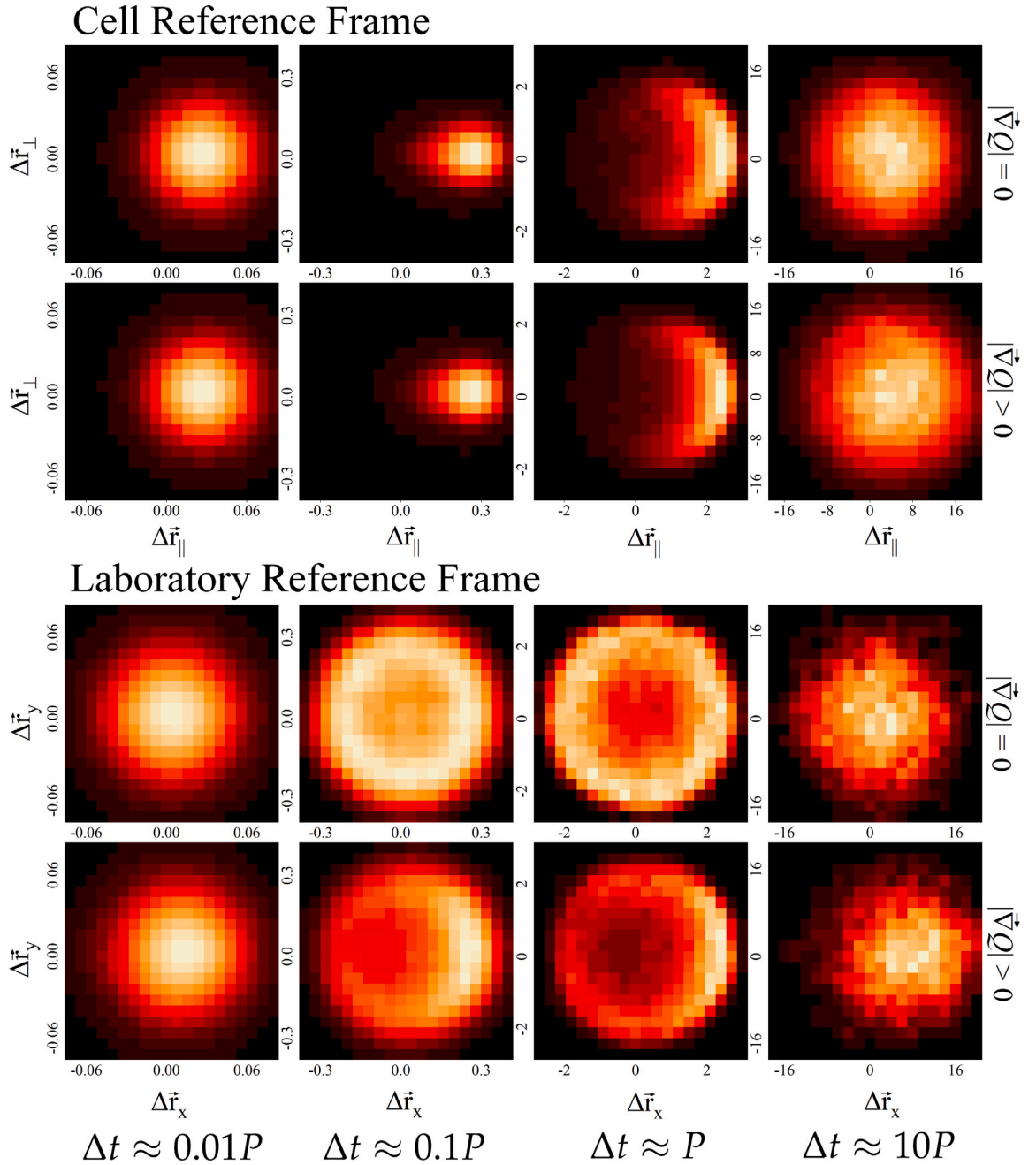


Fig. 11. The displacement distribution is the 2D histogram of the cell displacement in a given Δt . There are 4 columns of graphs, one for each $\Delta t \approx 0.01P, 0.1P, P, 10P$, where P is the persistence time. In the first 2 graph rows, the displacement is calculated in the cell reference frame defined by the cell polarization vector in the beginning of the step. The displacement relative to the polarization has coordinates $\Delta \vec{r}_{\parallel}$ (displacement parallel to polarization vector) and $\Delta \vec{r}_{\perp}$ (displacement perpendicular to polarization). In the 3rd and 4th graph rows, the displacement is calculated in the laboratory reference frame (Petry dish). The displacement relative to the Petry dish has coordinates $\Delta \vec{r}_x$ and $\Delta \vec{r}_y$. The cell parameters here are $\lambda_{F-actin} = 150$ and $\phi_f = 0.1$.

4.2. Takeaways for experimental biologists

Our model structure and data collection approach were inspired by the methodologies used in real-world experiments. The similarities enable us to make a few considerations on experiment design and data analysis regarding cell migration:

1. The sampling time interval and experiment length have to cover all expected regimes of cell movement. To determine if a time interval is small or large, run trials to check if the Mean Square Displacement — MSD encompasses all expected regimes.

As a starting point, the time it takes for a cell to make at least 2 successive polarization reorientations indicates a long-time interval; the life-time of a single lamellipodium protrusion indicates a short-time interval.

2. Investigate the presence of a diffusive regime for very short time intervals. Cells with frequent membrane fluctuation, lamellipodia protrusions and retractions, and interactions with other cells could present diffusion at short-time intervals. To check if the diffusive regime exists, plot the MSD in log-log scale and measure the MSD slope at short time intervals. A slope below 2 is a strong indicator of diffusive noise.
3. If cell movement presents a diffusive regime at short-time intervals, instantaneous velocity is not well defined. To circumvent this, use optimal velocity, which is the mean velocity measured for the time interval at which the MSD has its highest slope. To characterize cell motility, use drift speed, as this metric is well defined, provided the cell has polarization.
4. Fit the MSD curve. Besides the detection of different regimes, the MSD fit provides parameters such as diffusion, persistence, and terminal speed (when the cell responds to external cues). These parameters can be used to compare cell motility results from different experiments, and also with computational model results. We provide instructions to fitting an MSD curve with four regimes in [17]. The quality of the fit depends directly on whether all regimes of movement were covered (refer to item 1) and on the number of replicas (we used 10 replicas).
5. Using the cell reference frame to project metrics like cell displacement is a powerful technique to study the migration behavior of cells. For example, we demonstrated that our simulated cells have the same migration behavior regardless of the presence of external cues when analyzed at the cell reference frame, see Fig. 11.
6. Cell polarization unlocks the use of sophisticated metrics and circumvent the problems with instantaneous velocity. A proxy to polarization is any vector correlated to the optimal velocity: cell shape, average gradient of molecules related to cell migration signaling, average force exerted from the substrate, or the displacement of the nucleus from the rest of the cell body as we did in our simulations. Ref. [14] discusses how to determine the robustness of a polarization measurement protocol.

4.3. Final thoughts

We believe that our model can be applied to other taxis processes, since they are similar in a simplistic mathematical standpoint: the cell senses some anisotropy in the environment and adapts its migration direction. The simulation allows changes in the sensing mechanism, making it possible to study receptors' density, saturation, activation, delayed response, and forced protrusion. We are left with the task of building a pure mathematical model that can describe this specific dynamics, from which we should derive Eq. (3) with the extra ballistic term. We also aim at modifying this simulation to accomplish collective migration.

This model remains non validated, since we require datasets of cell trajectories with measurement of polarization. We hope that this paper underscores the significance of cell polarization measurement in the study of cell migration properties.

Our findings have a particular relevance to collective migration, as varying degrees of collective polarization and lamellipodium production can result in distinct behaviors. We thence expect that our paper will be useful for developing collective cell migration simulations in biological phenomena as in wound healing, metastasis and immune response.

CRediT authorship contribution statement

Pedro C. Dal-Castel: Writing – original draft, Visualization, Software, Investigation, Data curation. **Gilberto L. Thomas:** Writing – review & editing, Supervision, Software, Resources, Project administration, Methodology. **Gabriel C. Perrone:** Supervision, Software, Methodology, Conceptualization. **Rita M.C. de Almeida:** Writing – review & editing, Supervision, Project administration, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.physa.2025.130524>.

Data availability

Data will be made available on request.

References

- [1] Douglas A. Lauffenburger, Alan F. Horwitz, Cell migration: A physically integrated molecular process, *Cell* 84 (3) (1996) 359–369.
- [2] R. Fürth, Die brownische bewegung bei berücksichtigung einer persistenz der bewegungsrichtung. Mit anwendungen auf die bewegung lebender infusorien., *Z. Für Phys.* 2 (1920) 244–256.
- [3] Gilberto L Thomas, Ismael Fortuna, Gabriel C Perrone, Glazier James A, Julio M Belmonte, Rita M C de Almeida, Parameterizing cell movement when the instantaneous cell migration velocity is ill-defined, *Phys. A* 550 (2020) 124493.
- [4] Miguel Vicente-Manzanares, Alan Rick Horwitz, Cell migration: An overview, in: *Methods in Molecular Biology*, Humana Press, 2011, pp. 1–24.
- [5] Isabela C. Fortunato, Raimon Sunyer, The forces behind directed cell migration, *Biophysica* 2 (4) (2022) 548–563.
- [6] Alex Mogilner, Kinneret Keren, The shape of motile cells, *Curr. Biology* 19 (17) (2009) R762–R771.
- [7] Rita M C de Almeida, Guilherme S Giardini, Mendeli Vainstein, James A Glazier, Gilberto L Thomas, Exact solution for an anisotropic ornstein-uhlenbeck process, 2020, <http://arxiv.org/abs/2002.00051>.
- [8] Ismael Fortuna, Gabriel C Perrone, S Krug, Eduarda Susin, Julio M Belmonte, Gilberto L Thomas, Glazier James A, Rita M C de Almeida, CompuCell3D simulations reproduce mesenchymal cell migration on flat substrates, *Biophys. J.* 118 (11) (2020) 2801–2815.
- [9] Guilherme S.Y. Giardini, Gilberto L. Thomas, Carlo R. da Cunha, Rita M.C. de Almeida, Membrane fluctuations in migrating mesenchymal cells preclude instantaneous velocity definitions, *Phys. A* 647 (2024) 129915.
- [10] François Graner, James A. Glazier, Simulation of biological cell sorting using a two-dimensional extended potts model, *Phys. Rev. Lett.* 69 (1992) 2013–2016.
- [11] James A. Glazier, François Graner, Simulation of the differential adhesion driven rearrangement of biological cells, *Phys. Rev. E* 47 (1993) 2128–2154.
- [12] Paulien Hogeweg, Computing and organism: on the interface between informatic and dynamic proceses, *Biosystems* 64 (2002) 97–109.
- [13] Maciej Swat, Gilberto L. Thomas, Julio Monti Belmonte, Abbas Shirinifard, Dimitrij Hmeljak, James A. Glazier, Multi-scale modeling of tissues using CompuCell3D, *Methods Cell Biol.* 110 (2012) 325–366.
- [14] Gilberto L. Thomas, Ismael Fortuna, Gabriel C. Perrone, François Graner, Rita M.C. de Almeida, Shape-velocity correlation defines polarization in migrating cell simulations, *Phys. A* 587 (2022) 126511.
- [15] Włodzimierz Korohoda, Zbigniew Madeja, Jolanta Sroka, Diverse chemotactic responses of *Dictyostelium discoideum* amoebae in the developing (temporal) and stationary (spatial) concentration gradients of FolicAcid, cAMP, Ca^{2+} and Mg^{2+} , *Cell Motil. Cytoskeleton* 53 (2002) 1–25.
- [16] Matthew P. Neilson, Douwe M. Veltman, Peter J.M. van Haastert, Steven D. Webb, John A. Mackenzie, Robert H. Insall, Chemotaxis: A Feedback-Based Computational Model Robustly Predicts Multiple Aspects of Real Cell Behaviour, *PLoS Biol.* 9 (5) (2011) e1000618.
- [17] P.C. Dal-Castel, CompuCell3D model of cell migration reproduces chemotaxis - supplementary materials, 2024, <https://github.com/pdalc Castel/CC3D-Chemotaxis-SuppMat/tree/main>.
- [18] A.M. Turing, The chemical basis of morphogenesis, *Phil. Trans. R. Soc. Lond. B* 237 (1952) 37–72.
- [19] Kristen F. Swaney, Chuan-Hsiang Huang, Peter N. Devreotes, Eukaryotic chemotaxis: A network of signaling pathways controls motility, directional sensing, and polarity, *Annu. Rev. Biophys.* 39 (1) (2010) 265–289.
- [20] Orion D. Weiner, Paul O. Neilsen, Glenn D. Prestwich, Marc W. Kirschner, Lewis C. Cantley, Henry R. Bourne, A PtdInsP3- and rho GTPase-mediated positive feedback loop regulates neutrophil polarity, *Nature Cell Biol.* 4 (7) (2002) 509–513.
- [21] Chang Y. Chung, Richard A. Firtel, Signaling pathways at the leading edge of chemotaxing cells, *J. Muscle Res. Cell Motil.* 23 (7/8) (2002) 773–779.
- [22] Carol L. Manahan, Pablo A. Iglesias, Yu Long, Peter N. Devreotes, Chemoattractant signaling in dictyostelium discoideum, *Annu. Rev. Cell. Dev. Biol.* 20 (1) (2004) 223–253.
- [23] Evanthia T. Roussos, John S. Condeelis, Antonia Patsialou, Chemotaxis in cancer, *Nat. Rev. Cancer* 11 (8) (2011) 573–587.
- [24] Tian Jin, Xuehua Xu, Dale Hereld, Chemotaxis, chemokine receptors and human disease, *Cytokine* 44 (1) (2008) 1–8.
- [25] P.C. Dal-Castel, Single cell chemotaxis simulation CC3D, 2023, https://github.com/pdalc Castel/Single_Cell_Chemotaxis.2.3.
- [26] C. Metzner, F. Horsch, C. Mark, C. Voskens, B. Fabry, Detecting long-range interactions between migrating cells, *Sci Rep* 11 (2021) 15031.
- [27] Luke Tweedy, Börn Meier, Jürgen Stephan, Doris Heinrich, Robert G. Endres, Distinct cell shapes determine accurate chemotaxis, *Sci. Rep.* 3 (2606) (2013) 1–7.
- [28] Peter Devreotes, Chris Janetopoulos, Eukaryotic chemotaxis: Distinctions between directional sensing and polarization, *J. Biol. Chem.* 278 (23) (2003) 20445–20448.
- [29] Ru Cao, Yi Zhang, SUZ12 is required for both the histone methyltransferase activity and the silencing function of the EED-EZH2 complex, *Mol. Cell* 15 (1) (2004) 57–67.
- [30] Pere Roca-Cusachs, Raimon Sunyer, Xavier Trepat, Mechanical guidance of cell migration: lessons from chemotaxis, *Curr. Opin. Cell Biol.* 25 (5) (2013) 543–549.
- [31] Andrew C. Callan-Jones, Raphaël Voituriez, Actin flows in cell migration: from locomotion and polarity to trajectories, *Curr. Opin. Cell Biol.* 39 (2016) 12–17.
- [32] A.D. Viñales, M. Camuyrano, G.H. Paissan, Oscillations and negative velocity autocorrelation emerging from a Brownian particle model with hydrodynamic interactions, *Phys. Rev. E* 101 (2020) 052140, URL <https://link.aps.org/doi/10.1103/PhysRevE.101.052140>.
- [33] Fernando Peruani, Luis G. Morelli, Self-propelled particles with fluctuating speed and direction of motion in two dimensions, *Phys. Rev. Lett.* 99 (1) (2007) 010602.
- [34] Liang Li, Simon F. Nørrelykke, Edward C. Cox, Persistent cell motion in the absence of external signals: A search strategy for eukaryotic cells, *PLOS ONE* (2008) URL <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0002093>.
- [35] Peter Dieterich, Rainer Klages, Roland Preuss, Albrecht Schwab, Anomalous dynamics of cell migration, *PNAS* 105 (2008) 459–463.
- [36] Liang Li, Edward C. Cox, Henrik Flyvbjerg, ‘Dicty dynamics’: Dictyostelium motility as persistent random motion, *Phys. Biology* 8 (4) (2011) 046006.